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$f(x)=x \qquad g(x)=x^3$

$$|f(x)| = \sqrt{\int_{-1}^{1} x^2\,dx} \qquad |g(x)| = \sqrt{\int_{-1}^{1} x^6\,dx}$$

$$\frac{x^3}{3} = \left(\frac{2}{3}\right)\sqrt{\frac{2}{3}} \qquad \frac{x^7}{7} = \left(\frac{2}{7}\right)\sqrt{\frac{2}{7}} = \sqrt{\frac{4}{21}}$$

$\cos\alpha =$

↓

$$\cos\alpha = \frac{\sqrt{21}}{2} \int_{-1}^{1} f(x)g(x)$$

$$\frac{\sqrt{21}}{2} \int_{-1}^{1} x^4$$

$$\frac{x^5}{5}$$

$$\frac{1}{5} - \left(\frac{-1}{5}\right) = \frac{2}{5}$$

$$\frac{\sqrt{21}}{2} \cdot \frac{2}{5} = \frac{\sqrt{21}}{5}$$

$$\cos\alpha = \frac{\sqrt{21}}{5}$$

CamScanner ile tarandı